



January 15, 2019 (Sent by e-mail)

Director of Waste Management  
c/o Liliana Jerade  
BC Ministry of Environment and Climate Change Strategy  
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Attention: Liliana Jerade, Senior Contaminated Sites Officer

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**Re: Q3/Q4 2018 Progress Report for Management of High Risk Conditions at the 5100 Portal Area of the former Sunro Mine, Jordan River BC, Site ID 16790**

This letter provides a semi-annual update for the 5100 Portal Area of the former Sunro Mine ("Site"), as requested in the Ministry of Environment and Climate Change Strategy's (ENV) letter to Teck Resources Limited (Teck) and Western Forest Products Inc. (WFP), dated September 18, 2017.

Since the last progress update (Q1/Q2 2018), Teck submitted the SNC-Lavalin Inc. report titled *Supplemental Detailed Site Investigation, Sunro Mine Jordan River, BC* dated July 27<sup>th</sup>, 2018; executed a supplemental site investigation and continued with additional studies to support the aquatic ecological risk assessment (ERA) on the Jordan River. The purpose of this work was to collect aquatic biological and baseline water quality and quantity data to inform risk management and remediation planning for mine wastes at the Site. Work completed as part of the supplemental site investigation and ERA included:

- Completion of summer and fall sampling of water quality and quantity to support baseline (i.e., "pre-remediation") data, assess temporal estimates of metals inputs to the river and potential effects from seasonality or river flow rates. The assessment included collecting surface water, porewater and groundwater samples across the Site and within the adjacent Jordan River. The assessment included characterizing water quality and quantity discharging from the 5100 Portal.
- Delineation of hydrocarbon exceedances of the Contaminated Site Regulation (CSR) standard protective of aquatic life observed in samples collected from monitoring well MW18-05. The samples collected from the nearest downgradient monitoring well (MW18-06) showed detectable xylenes and LEPH but at concentration less than the aquatic life standards.
- Azimuth and D. Burt and Associates conducted a field study targeting salmonids in the Jordan River in September as a follow up on the initial findings of the copper bioavailability assessment (i.e., using the biotic ligand model). Key components of the study were a fish population survey and fish gill copper sampling. The population survey was a repeat of a series of investigations conducted in support of BC Hydro's Water Use Planning where fish abundance, biomass and condition factor were compared between upstream (reference) stations and downstream (exposure) stations. In addition, gill tissue samples were also collected to help link the field survey to the biotic ligand modelling. Results of all these studies are currently being interpreted.

Work in progress or planned in Q1/Q2 in 2019, includes:

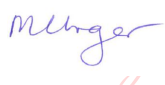
- Collection of baseline water data from the Site in January focussing on the groundwater to porewater and surface water pathways and portal discharge as part of the site investigation;
- Refinement of a seasonal loading model using the groundwater data;
- Reporting on the supplemental site investigation and aquatic ecological risk assessment;

- Development of preferred remedial options based on review of the site investigation and ERA reports;
- Information sharing and engagement of conceptual remedial options with the two local First Nation groups, Jordan River Round Table and government officials; and
- Evaluation of preferred remediation option(s).
- Continued team meetings to ensure remediation at the Site occurs in a timely manner.

The above activities will provide the necessary information to advance to the next step of remediation planning. We anticipate that selection of a preferred plan and initiation of permitting would occur in 2019, with implementation of remediation to occur in 2020.

We trust this meets your current requirements. Please contact me if you have questions.

Yours sincerely,

 Digitally signed by Michelle Unger  
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